

LOUIS SUNGWOO CHO

ITS & Smart Mobility Engineer | Illinois E.I.T

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PROFESSIONAL SUMMARY

Intelligent Transportation Systems (ITS) Engineer with hands-on experience with Miovision traffic management and analytics tools, Chicago Department of Transportation (CDOT) traffic operations, Advanced Transportation Management Systems (ATMS), Transit Signal Priority (TSP), and Python/SQL-based transportation data analysis. Experienced using Miovision Traffic Insights, camera and device monitoring, telemetry, signal event monitoring, and traffic data analysis to investigate operational conditions and data-quality issues. Background in connected and automated vehicle research, ROS-based autonomous systems, traffic-flow modeling, and rapid technical prototyping. Skilled at working across agencies, consultants, vendors, and engineering teams to diagnose technical problems, analyze system data, communicate findings, and translate transportation needs into practical ITS solutions.

EDUCATION & CERTIFICATIONS

M.S. Civil Engineering (Transportation)

University of Illinois Urbana-Champaign | May 2025

B.S. Honors Civil Engineering, Minor in Computer Science

University of Illinois Urbana-Champaign | May 2024

Illinois E.I. T

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CORE SKILLS

ITS/Traffic: Advanced Transportation Management Systems (ATMS); Transit Signal Priority (TSP); traffic operations; signal timing datasets; corridor performance monitoring; real-time traffic monitoring at the CDOT Traffic Management Center

Technical Communication: Technical summaries; system architecture documentation; ITS project reporting; presentation-ready technical content; stakeholder presentations

Solutions Engineering: Technical discovery; agency/customer requirements; rapid prototyping; hardware/software integration; proof-of-concept validation; technical demonstrations; cross-functional problem solving

Data Analytics: Python and SQL for traffic-volume analysis and cleaning; database QA/QC; R-Shiny dashboard support; data reliability reporting; Large Language Models

ITS Platforms: Miovision and ATMS camera and device monitoring; telemetry monitoring; signal event monitoring; traffic volume analysis; operational troubleshooting and QA/QC

Project Coordination: CDOT, CTA, vendor, and consultant coordination; procurement workflow support; deliverable tracking; stakeholder communication; cross-functional project support

PROFESSIONAL EXPERIENCE

ITS Consultant | Alrek, Contract with Chicago Department of Transportation (CDOT) | Chicago, IL

Dec 2025 - Present

- Support CDOT, Miovision, Parsons, and AECOM on ITS and traffic operations initiatives involving ATMS, Chicago Traffic Tracker, traffic data, signal timing datasets, Transit Signal Priority, and corridor performance monitoring.
- Leverage Miovision cameras, device telemetry, signal event monitoring, and traffic analytics to evaluate operational conditions and data quality; perform QA/QC of the Miovision Mateo feature and validate extracted traffic data.
- Diagnose traffic-volume discrepancies across Miovision, R-Shiny, and CDOT data sources in collaboration with Parsons engineers and agency stakeholders.
- Perform Python and SQL-based QA/QC of CDOT traffic databases to identify duplicate, missing, and inconsistent traffic-count records and support operational decision-making.
- Support CDOT/CTA Transit Signal Priority initiatives through signal timing data preparation, procurement coordination, and cross-agency technical collaboration.
- Coordinate the Chicago Accessible Intersection Navigation (CAIN) project across vendors and transportation engineers; developed and presented the technical project poster to USDOT SMART Grant stakeholders at ITS America 2026 Conference at Detroit, Michigan.
- Monitor real-time traffic operations from the CDOT Traffic Management Center and communicate operational findings to project teams.

Autonomous Vehicles Research Assistant | University of Illinois Urbana-Champaign | Urbana, IL

May 2022 - May 2025

- Developed and validated ROS and Python-based planning and control functionality on a physical autonomous vehicle platform at the Illinois Center for Transportation.
- Applied traffic-flow modeling and stability analysis to mixed human-autonomous traffic and analyzed robotaxi trajectories for lane-changing and car-following behavior.
- Conducted trajectory-prediction research using deep learning and implemented collision-avoidance and vehicle-control algorithms in simulation and physical testing environments.

SELECTED PROJECTS

- **CAIN Pedestrian Accessibility ITS Project** – Developed and presented a technical ITS poster at ITS America 2026 highlighting accessible pedestrian navigation, system architecture, stakeholder coordination, and smart-mobility applications.
- **Miovision Devices & Parsons ATMS QA/QC** – Evaluated ATMS, Chicago Traffic Tracker, and Miovision traffic data to identify data-quality discrepancies, validate traffic conditions, and support corridor operations analysis.
- **Chicago Centralized Transit Signal Priority (TSP)** – Supported CDOT/CTA TSP implementation through signal timing data preparation, procurement coordination, technical documentation, and cross-agency stakeholder alignment.